
Q1. Explain Security Goals

Security goals (also called the **CIA Triad**) ensure protection of information systems:

- **Confidentiality** – Ensures that data is accessible only to authorized users.
 - **Integrity** – Ensures that data is accurate and not altered by unauthorized users.
 - **Availability** – Ensures that systems and data are available when required.
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Q2. Define Active and Passive Attacks

Active Attack:

An attack in which the attacker modifies, disrupts, or damages the system or data.

Example: Data modification, denial of service

Passive Attack:

An attack in which the attacker only monitors or listens to data without altering it.

Example: Eavesdropping, traffic analysis

Q3. Discuss Goals of Security

The main goals of security are:

- **Confidentiality** – Protects sensitive information from unauthorized access.
 - **Integrity** – Maintains accuracy and consistency of data.
 - **Availability** – Ensures timely access to data and resources.
 - **Authentication** – Verifies the identity of users or systems.
 - **Authorization** – Grants permissions to access resources.
 - **Non-repudiation** – Prevents denial of actions or transactions.
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Q4. Euclidean Algorithm

The **Euclidean Algorithm** is a method used to find the **Greatest Common Divisor (GCD)** of two integers by repeatedly applying division.

Formula / Concept

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

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Step 1: Start

Step 2: Input a, b

Step 3: While b $\neq$ 0 do

 r = a mod b

 a = b

 b = r

Step 4: End While

Step 5: Output a as GCD

Step 6: Stop

Steps of Algorithm

1. Divide the larger number by the smaller number.
 2. Replace the larger number with the smaller number.
 3. Replace the smaller number with the remainder.
 4. Repeat the process until the remainder becomes **0**.
 5. The last non-zero remainder is the **GCD**.
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Example

Find GCD of 48 and 18:

- $48 \div 18 = 2$ remainder **12**
- $18 \div 12 = 1$ remainder **6**
- $12 \div 6 = 2$ remainder **0**

GCD = 6

Advantages

- Efficient and fast method
 - Works for large numbers
 - Widely used in cryptography
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Q5.Substitution and Transposition Techniques

1. Substitution Technique

Substitution is a method of encryption in which each element (letter/character) of the plaintext is **replaced by another character** according to a specific rule.

- The **order of characters remains the same**
- Only the **characters are changed**
- Used in classical ciphers

Examples:

- Caesar Cipher
- Monoalphabetic Cipher
- Vigenère Cipher

Example:

Plaintext: HELLO

Ciphertext (shift by 3): KHOOR

2. Transposition Technique

Transposition is a method of encryption in which the **positions of characters are changed**, but the characters themselves remain the same.

- The **characters are not altered**
- Only their **order is rearranged**
- Makes the message difficult to read

Examples:

- Rail Fence Cipher
- Columnar Transposition Cipher

Example:

Plaintext: HELLO

Ciphertext (rearranged): LEOHL

Key Difference

- **Substitution:** Changes characters but keeps order same
 - **Transposition:** Changes order but keeps characters same
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Q6.Vigenère Cipher

The **Vigenère Cipher** is a method of encryption that uses a **keyword to apply multiple Caesar shifts** on the plaintext, making it more secure than simple substitution ciphers.

Concept

- It is a **polyalphabetic substitution cipher**
 - Each letter of plaintext is shifted using a corresponding letter of the **repeating keyword**
 - Uses a **Vigenère table (tabula recta)**
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Encryption Formula

$$C_i = (P_i + K_i) \bmod 26$$

Where:

- P_i = Plaintext letter value
- K_i = Key letter value
- C_i = Ciphertext letter

Steps for Encryption

1. Write the plaintext.
2. Repeat the keyword to match the length of plaintext.
3. Convert letters to numbers (A=0, B=1, ..., Z=25).
4. Apply the formula to get ciphertext.

Example

Plaintext: ATTACKATDAWN

Keyword: LEMONLEMONLE

Ciphertext: LXFOPVEFRNHR

Decryption Formula

Decryption Formula

$$P_i = (C_i - K_i) \bmod 26$$

Advantages

- More secure than Caesar cipher

- Difficult to break using frequency analysis
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Disadvantages

- Repeating key makes it vulnerable to attacks
 - Can be broken using Kasiski examination
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